

* karyosis

4 12, 14, 15, 16, 18, 20, 22, 24, 26, 100

1 45, 46, 47, 48, 49, 50, 51, 52, 53, 120

x	$x - \text{Mean}$	$(x - \bar{x})^2$	$(x - \bar{x})^4$
45	-11.1	123.21	15180.704
46	-10.1	102.01	10406.040
47	-9.1	82.81	6857.496
48	-8.1	65.61	4304.672
49	-7.1	50.41	2541.168
50	-6.1	37.21	1384.584
51	-5.1	26.01	676.52
52	-4.1	16.81	282.58
53	-3.1	9.61	92.04
120	63.9	4083.21	16672.003

$$\bar{x} = \frac{561}{10} = 56.1 \quad \sum x^2 = 4596.9$$

$$\sum (x - \bar{x})^4 = 16,714,200$$

② variance =

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{4596.9}{10}$$

$$= 459.69$$

(3) standard deviation

$$s^2 = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$= \sqrt{459.69}$$

$$= \sqrt{21.4}$$

(4) kurtosis

$$k = \frac{\sum \frac{(x - \bar{x})^4}{N \cdot s^4}}$$

$$s^4 = (21.4)^2$$

$$= 209727.36$$

$$459.67 \times 459.67$$

$$= 211296.50$$

$$\frac{16672563.83}{10.211296.50}$$

$$\frac{16672563.83}{21,12,965.1}$$

$$\boxed{7.89}$$

$$\boxed{7.89 > 3}$$